ZF plus the existence of a Berkeley cardinal

First-order axiomatization for the Berkeley1399 verification package

Language and conventions. Classical first-order logic with equality and the sole nonlogical symbol $\in$. Take universal closures of the displayed axioms. All sets, functions, ordinals, formula codes, and truth relations below are internal to this set theory. Neither Choice nor Dependent Choice is added.

ZF axioms. In the two schemas, formulas may have arbitrary finite tuples of parameters $\bar{p}$; bound and auxiliary variables are chosen fresh. Write $x \subseteq a$ for $\forall t(t \in x \to t \in a)$ and $x^+ = x \cup \{x\}$.

$$
\begin{aligned}
\text{Extensionality:}\quad & [\forall x(x \in a \leftrightarrow x \in b)] \to a = b.\\
\text{Empty set:}\quad & \exists e\, \forall x\ (x \notin e).\\
\text{Pairing:}\quad & \exists c\, \forall x\ [x \in c \leftrightarrow (x = a \vee x = b)].\\
\text{Union:}\quad & \exists b\, \forall x\ [x \in b \leftrightarrow \exists y \in a\ (x \in y)].\\
\text{Power set:}\quad & \exists b\, \forall x\ [x \in b \leftrightarrow x \subseteq a].\\
\text{Infinity:}\quad & \exists a\ [\varnothing \in a\ \wedge\ \forall x \in a\ (x^+ \in a)].\\
\text{Separation:}\quad & \exists b\, \forall x\ [x \in b \leftrightarrow (x \in a \wedge \varphi(x, \bar{p}))], \quad b \notin \mathrm{FV}(\varphi).\\
\text{Replacement:}\quad & [\forall x \in a\ \forall y\, \forall z\ ((\varphi(x, y, \bar{p}) \wedge \varphi(x, z, \bar{p})) \to y = z)]\\
& \qquad \to \exists b\, \forall y\ [y \in b \leftrightarrow \exists x \in a\ \varphi(x, y, \bar{p})], \quad b \notin \mathrm{FV}(\varphi).\\
\text{Foundation:}\quad & a \neq \varnothing\ \to\ \exists x \in a\ \forall y \in x\ (y \notin a).
\end{aligned}
$$

Replacement is the bounded-domain, partial-functional form: each input has *at most* one output. Over the other ZF axioms it is equivalent to the usual total-functional schema.

Set-coded syntax and truth. Use Kuratowski pairs $\langle a, b\rangle = \{\{a\}, \{a, b\}\}$ and the least inductive set ω. Let $\mathcal{F}$ be the least set closed under the following constructors ($i, k \in \omega$ and $p, q \in \mathcal{F}$):

$$
\mathsf{eq}(i,k) = \langle 0, \langle i, k\rangle\rangle, \quad \mathsf{mem}(i,k) = \langle 1, \langle i, k\rangle\rangle, \quad \mathsf{neg}(p) = \langle 2, p\rangle, \quad \mathsf{imp}(p,q) = \langle 3, \langle p, q\rangle\rangle, \quad \mathsf{all}(i,p) = \langle 4, \langle i, p\rangle\rangle.
$$

For a set M, let E_M be the set of all function graphs $s : \omega \to M$. The first-order condition $\mathsf{Truth}_M(S)$ says $S \subseteq \mathcal{F} \times E_M$ and the following five clauses hold for all indicated codes and assignments; abbreviate $\langle p, s\rangle \in S$ by $S(p, s)$:

$$
\begin{aligned}
& S(\mathsf{eq}(i,k), s) \leftrightarrow s(i) = s(k), && S(\mathsf{mem}(i,k), s) \leftrightarrow s(i) \in s(k),\\
& S(\mathsf{neg}(p), s) \leftrightarrow \neg S(p, s), && S(\mathsf{imp}(p,q), s) \leftrightarrow (S(p, s) \to S(q, s)),\\
& S(\mathsf{all}(i,p), s) \leftrightarrow \forall x \in M\ S(p, s[i \mapsto x]).
\end{aligned}
$$

Here $s[i \mapsto x]$ changes just coordinate i. Least closure, function graphs, evaluation, and updates are definable using $\in$ and $=$; ZF proves the required sets and truth relation exist.

The Berkeley axiom B. Put $\mathrm{Trans}(M) \equiv \forall x \in M\ \forall y \in x\ (y \in M)$. $\mathrm{Ord}(\delta)$ means $\mathrm{Trans}(\delta)$ and $\in$ is a strict total order on δ (Foundation supplies well-foundedness). $\mathrm{Card}(\delta)$ means $\mathrm{Ord}(\delta)$ and there is no bijection from any $\alpha \in \delta$ onto δ. The additional axiom is the single sentence

$$
\begin{aligned}
&\exists \delta\ \big[\mathrm{Card}(\delta) \wedge \omega \in \delta\ \wedge\ \forall M\big((\mathrm{Trans}(M) \wedge \delta \in M)\ \to\ \exists S\,[\mathsf{Truth}_M(S)\ \wedge\\
&\forall \eta \in \delta\ \exists j\, \exists \kappa\,[j : M \to M\ \wedge\ \eta \in \kappa \wedge \kappa \in \delta \wedge j(\kappa) \neq \kappa \wedge (\forall \alpha \in \kappa\ (j(\alpha) = \alpha))\\
&\qquad\qquad \wedge\ (\forall p \in \mathcal{F}\ \forall s \in E_M\ (S(p, s) \leftrightarrow S(p, j \circ s)))],]\big)\big].
\end{aligned}
$$

Thus $j : (M, \in) \to (M, \in)$ is elementary and $\eta < \mathrm{crit}(j) < \delta$. One j preserves *all* internally coded formulas. This is a first-order sentence via the set-coded definitions above, not a separate axiom for each formula.

Formal correspondence. The package proves halting iff finite contradictory derivations in `BoundedReplacementND`; `TheoryTheorem.v` also gives the model identification with internal all-code Berkeley elementarity. The exact expanded sentence is `machine/berkeley-pure.sexpr`; the source axiom schemes are `machine/axioms.sexpr`. See `docs/proof-overview.md` for the precise metatheoretic boundary and variable conventions.

Reference: Bagaria, Koellner, and Woodin, *Large Cardinals Beyond Choice*, §3.2, Definition 3.4.